

Scientific Paper Categorization + Extractive Summary

Problem:

Automatically categorize scientific research papers into fields (e.g., CS, physics, math, etc.) and generate an **extractive summary** that highlights the most important sentences.

Dataset:

arXiv Metadata Dataset

Step-by-Step Plan

1. Define Objective & Scope

- Task 1: Categorize research papers into scientific fields.
- Task 2: Generate an **extractive summary** using NLP algorithms.
- Output: An API where a user uploads a paper → returns **category + summary**.

2. Data Collection & Preprocessing

- Use arXiv dataset (title, abstract, categories, etc.).
- Preprocess text: tokenization, lowercasing, stopword removal.
- Convert dataset into:
 - **Classification dataset** (abstract → category).
 - **Summarization dataset** (abstract/full text → key sentences).

3. Baseline Models

- Classification: **TF-IDF + Naive Bayes**.
- Summarization: **TextRank** for extractive summaries.
- Establish benchmark accuracy and ROUGE scores.

4. Deep Learning Models

- Classification: **CNN for text classification**.
- Compare with: LSTM or BiLSTM (sequence-based).
- Summarization: Keep **TextRank** for extractive (no Transformers).

5. Model Training & Evaluation

- Train models on labeled arXiv dataset.
- Metrics:
 - Classification → Accuracy, F1-score.
 - Summarization → ROUGE, BLEU.
- Save trained models.

6. Similarity & Explainability (Optional Add-on)

- Use cosine similarity to link papers to closest field clusters.
- Provide **top-3 related categories** instead of only one.

7. MLOps with MLflow

- Track models (baseline vs CNN vs LSTM).
- Log hyperparameters, metrics, artifacts (confusion matrix, ROUGE scores).
- Compare classification models directly.

8. Automation with Airflow

- DAG for:
 1. Ingest new arXiv papers.
 2. Preprocess text.
 3. Run classification + summarization.
 4. Update results in database/API.

9. Deployment with Docker + Kubernetes

- Package model into a **REST API** (FastAPI).
- Input: research paper text.
- Output: {category: "CS", summary: "...summary..."}
- Deploy as scalable microservice on Kubernetes.

10. Final Deliverables

- **API Service:** Upload a paper → get category + summary.
- **Dashboard:** View paper categories, summaries, and evaluation metrics.
- **MLOps Tracking:** MLflow experiments, Airflow pipelines.

